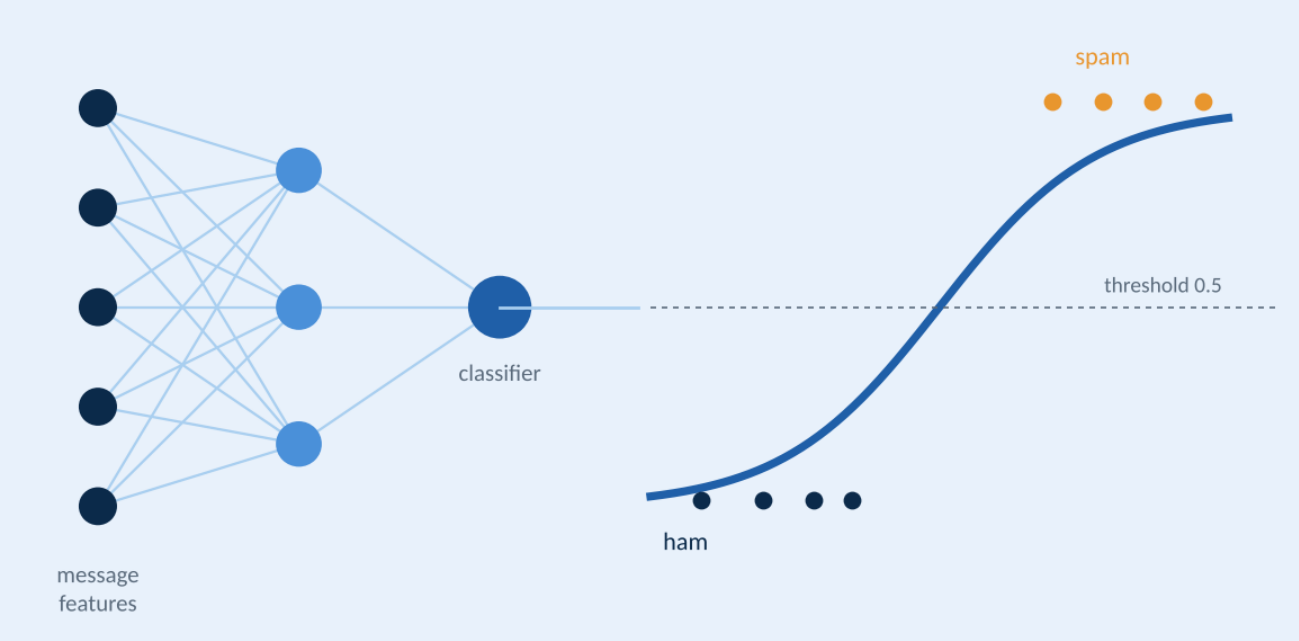
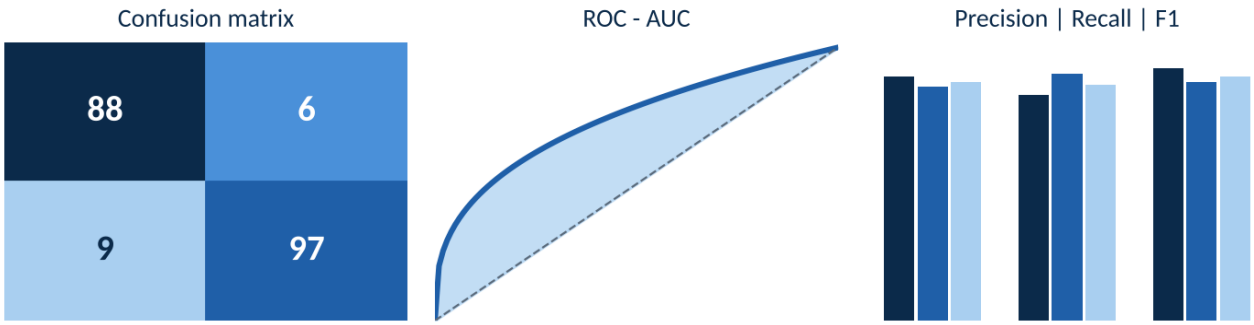
SUPERVISED MACHINE LEARNING

MESSAGE

INTELLIGENCE SYSTEM

Part A: Probability & Conceptual Foundation (Theory)





Supervised Learning | Spam & ham message classification

Part A: Probability & Conceptual Foundation (Theory)

Q1. What is Conditional Probability?

Conditional probability is the probability that an event **A** occurs given that another event **B** has already occurred. It is written **P(A|B) = P(A∩B) / P(B)**, with P(B) > 0. Knowing B shrinks the sample space to B alone, so we only count the part of A that falls inside it.

In our project: the chance a message is spam changes once we see that it contains the word "free".

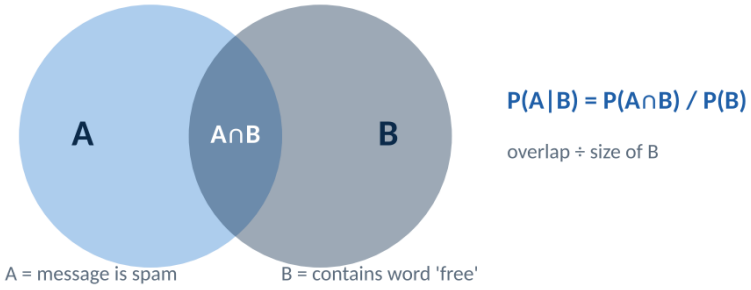


Fig 1. Conditional probability as the overlap A∩B within B

Q2. Explain Bayes' Theorem and its importance in classification problems.

Bayes' Theorem reverses a conditional probability: **P(C|x) = P(x|C) · P(C) / P(x)**. It updates our prior belief about a class **C** using the evidence **x** that we observe, and gives the posterior probability.

- **Why it matters:** P(x|C) is easy to estimate from labelled data, while P(C|x) is what we actually need.
- The classifier simply predicts the class with the **highest posterior**; priors also handle imbalanced classes.
- **Example:** P(spam)=0.30, P("free"|spam)=0.60, P("free"|ham)=0.05 gives P(spam|"free") = 0.18 / (0.18+0.035) ≈ **0.84**.

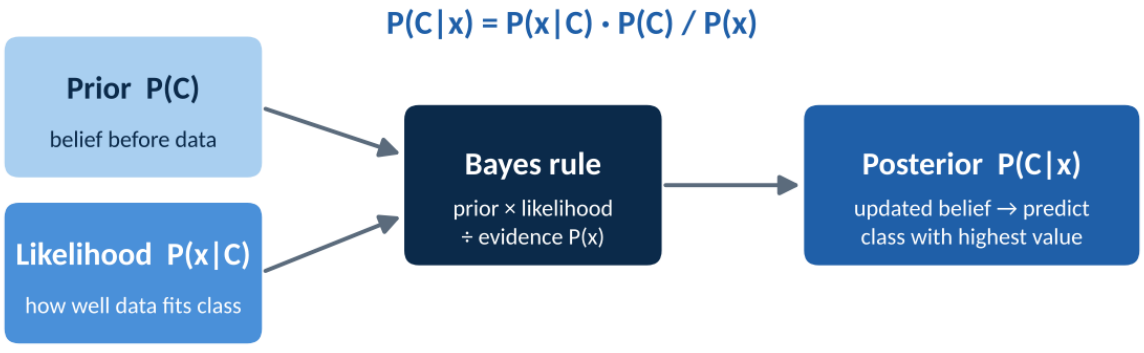


Fig 2. Prior and likelihood combine into the posterior

Q3. What assumptions does the Naive Bayes Classifier make?

Naive Bayes applies Bayes' rule with a strong simplifying assumption, which gives $P(C|x) \propto P(C) \cdot \prod P(x_i|C)$. Its main assumptions are:

- **Conditional independence:** given the class, every feature (word) is independent of the others.
- **Equal contribution:** each feature is treated as equally important to the outcome.
- **Known distribution:** features follow an assumed form (Gaussian, Multinomial or Bernoulli).
- **Representative data:** training frequencies reflect real priors; Laplace smoothing avoids zero probabilities.

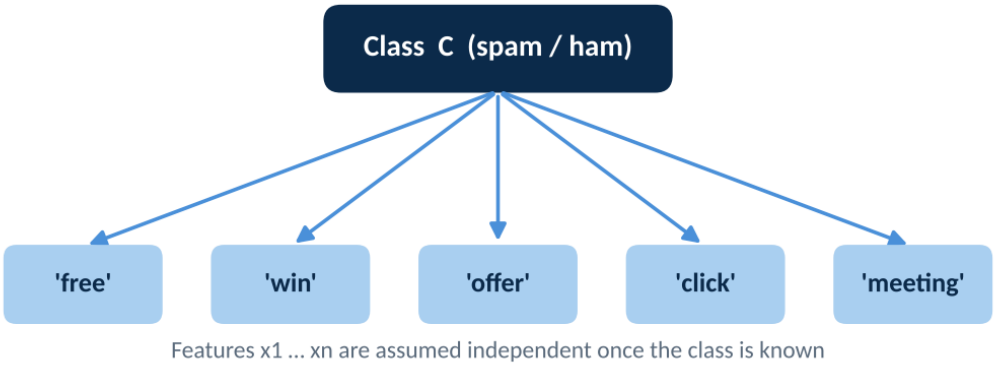


Fig 3. Naive Bayes: features depend only on the class

Q4. Explain the working principle of KNN and SVM.

K-Nearest Neighbors (KNN) is a lazy, instance-based method. It stores all training data and does no real training.

- For a new message, measure its distance (usually Euclidean) to every training point.
- Pick the K closest points and assign the class chosen by majority vote.
- A small K is noisy, a large K is over-smoothed; features must be scaled first.

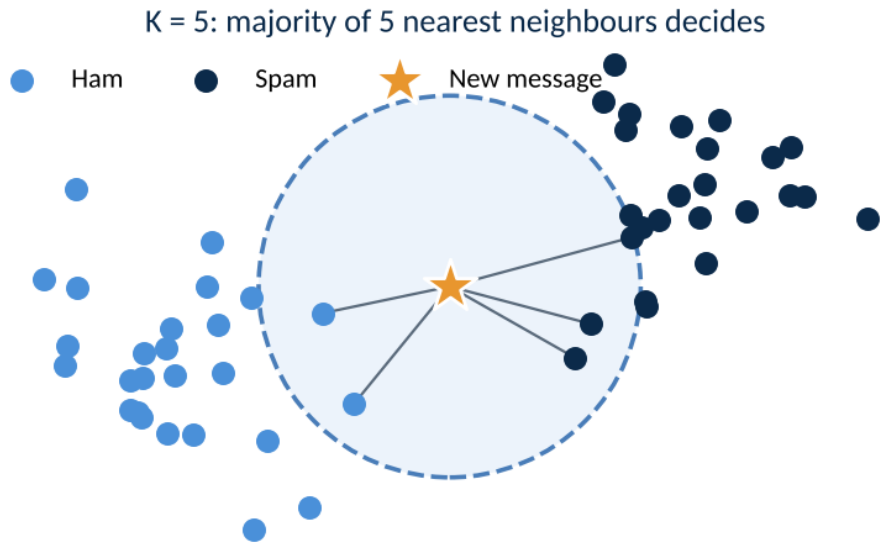


Fig 4. KNN with K = 5

Support Vector Machine (SVM) finds the hyperplane that separates two classes with the **maximum margin**.

- Only the points closest to the boundary, the **support vectors**, define the hyperplane.
- A soft margin (parameter **C**) tolerates a few misclassified points.
- The **kernel trick** (RBF, polynomial) maps data to a higher dimension so non-linear classes can be separated.

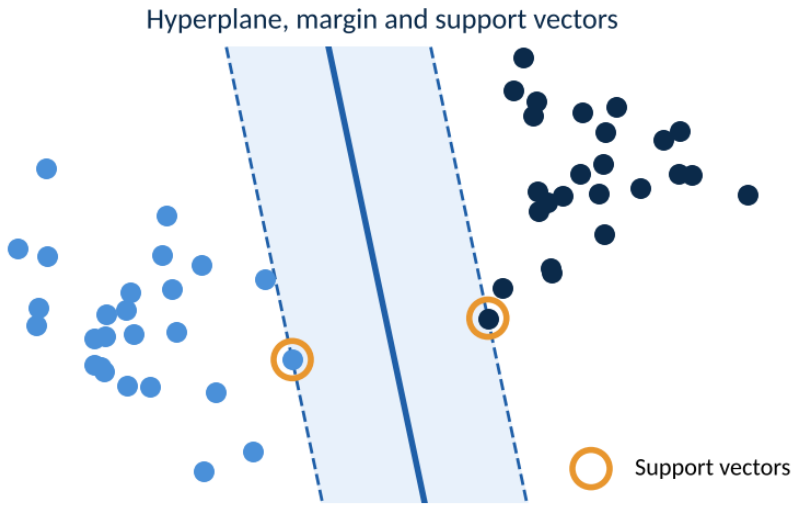


Fig 5. SVM: maximum-margin hyperplane and support vectors

Q5. Compare distance-based, probabilistic, and margin-based classifiers.

The three families differ in how they reach a decision: by closeness, by probability, or by geometry.

	Distance-based (KNN)	Probabilistic (Naive Bayes)	Margin-based (SVM)
Core idea	Classify by similarity to stored examples	Classify by the most probable class using Bayes' rule	Classify using the widest-margin boundary between classes
Learns	Nothing (lazy learner); keeps the data	Class priors and feature likelihoods	Support vectors and a hyperplane
Decision rule	Majority vote of K nearest points	Class with highest posterior P(C|x)	Sign of w·x + b
Strengths	Simple, no training, handles non-linear shapes	Very fast, works with little data, ideal for text	Strong on high-dimensional data, good generalisation
Weaknesses	Slow on large data, needs feature scaling	Independence assumption is rarely true	Slower to train, needs tuning of C and kernel
In this project	Find messages similar to a new one	Spam score from word frequencies	Separate spam from ham in TF-IDF space

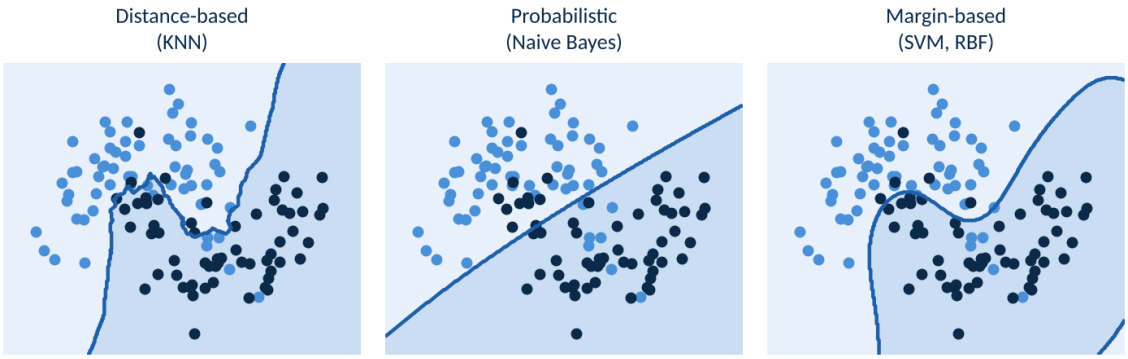


Fig 6. Typical decision boundaries of the three families